

PROJECT PROPOSAL

GROUP PROJECT

IIT372-2

Group No: IIT -11

**SMART FARM MANAGEMENT AND
ADVISORY SYSTEM**

Industrial Information Technology

Department of Computer Science and Informatics

Faculty of Applied Sciences

Uva Wellassa University of Sri Lanka

2026

Declaration

We hereby declare that the project will be developed by us and will be our own effort and that no part will be plagiarized without citations under the supervision of Ms. J. Jenusiya and co supervisor Mr. M. Janotheepan and Ms. L. Mathumila. This Project Proposal is submitted for the partial fulfillment of the requirement of the course unit IIT372-2, Project II for the degree of Computer Science and Technology.

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Group No: IIT _ 11

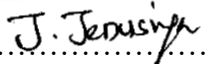
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Chapter 1 : Introduction

1.1 Project Title

Smart Farm Management and Advisory System

1.2 Project Description

The Smart Farm Management and Advisory System is a web-based agricultural management platform developed to support modern farming practices for farmers and farm managers. The system is designed to improve productivity, efficiency, and decision making by integrating multiple farming operations into a single digital solution. Its main focus is crop management, including crop growth monitoring, field activity management, irrigation scheduling, pest control, and farm progress tracking, where farmers can update daily agricultural activities based on field operations and system accessibility. Farm managers can supervise and manage overall farm operations in real time. The system also includes a supporting livestock management module for feeding records, health monitoring, and production tracking in mixed farming environments.

In addition, the system provides AI-based advisory help farmers make informed decisions. These features include smart farming recommendations based on weather conditions and farm data. The platform also includes an agricultural marketplace where farmers can directly sell crops and livestock products to customers, along with QR-based verification for transparency and traceability. Overall, the system combines crop management, livestock support, advisory services, and marketplace functionality to improve farm efficiency and agricultural decision-making.

1.3 Background and Motivation

Effective farm management is essential for improving agricultural productivity, ensuring efficient resource utilization, and supporting better decision-making in both crop and livestock farming. However, many small and medium-scale farmers still rely on traditional manual methods such as notebooks, spreadsheets, and verbal communication to manage daily farm activities. These methods often result in poor record keeping, lack of coordination, and delays in decision-making, which negatively affect overall farm performance.

In most farming environments, important information such as crop growth stages, irrigation schedules, livestock feeding records, and production data is not systematically recorded or monitored. This creates difficulties in tracking farm activities accurately and reduces the ability of farmers to respond quickly to issues such as pests, diseases, or production losses. In addition, access to reliable advisory information regarding weather conditions, market prices, and agricultural practices remains limited for many farmers [1].

With the advancement of web-based technologies, agriculture can now be improved through centralized digital systems that allow efficient data storage, real-time updates, and easy access to information. Web applications enable farmers and farm managers to manage operations more effectively by reducing dependency on manual processes and improving accuracy in record keeping. These systems also support better organization and monitoring of farm activities in a structured manner [2].

However, most existing systems do not provide a fully integrated solution that combines crop management, livestock support, advisory services, and marketplace functionality within a single platform. Therefore, the motivation for this project is to develop a simple and integrated web-based solution that improves traditional farm management practices. The Smart Farm Management and Advisory System focus on crop management while including livestock as a supporting module, along with marketplace and advisory features to enhance productivity and decision-making.

1.4 Problem in Brief

Effective farm management is essential for ensuring high productivity, efficient resource utilization, and better decision-making in agricultural activities [1]. However, many small to medium-scale farms still rely on traditional and fragmented methods, leading to several operational challenges.

- Lack of Digital Crop Tracking and Record Management

Crop growth stages and farm activities are not systematically recorded, and farmers rely on manual methods such as notebooks or spreadsheets. This results in data loss, inconsistencies, and difficulty in monitoring farm performance [1].

- Limited Decision Support and Resource Planning

Farmers have limited access to real-time insights such as weather-based recommendations in traditional farming environments. This can lead to inefficient use of resources such as water, fertilizers, and feed, increasing the risk of crop failure and wastage [2].

- Insufficient Livestock and Workforce Management System

There is no proper system to monitor livestock health, feeding schedules, or production records. In addition, farm workers are not efficiently managed in terms of task assignment, attendance, and salary tracking, leading to poor coordination and reduced productivity [1].

- Limited Market Access and Product Transparency

Farmers face challenges in directly selling their products to customers. Additionally, there is no proper system to verify product origin and quality, which reduces customer trust and market transparency [1].

These challenges lead to inefficiencies in farm operations, poor decision-making, and reduced profitability. Therefore, there is a clear need for a smart, integrated system that can manage all core farming activities in a unified platform.

1.5 Proposed Solution

To address the challenges in traditional farm management, the Smart Farm Management and Advisory System is proposed as a centralized web-based platform that integrates crop management, livestock support, decision-making tools, and market access into a single system.

The platform introduces the following key innovations:

- **Centralized Farm Management System**
Provides a unified platform to manage crop activities, livestock data, workforce, and farm records for better coordination and consistency.
- **Structured Crop Tracking System**
Enables real-time monitoring of crop growth stages and field activities from planting to harvest, improving planning and progress tracking.
- **AI-Based Decision Support and Advisory System**
Provides smart recommendations based on weather and farm data for irrigation, crop care, and resource use. It also includes crop disease detection for early identification and timely treatment.
- **Livestock Management Support Module**
Supports basic monitoring of livestock activities such as feeding, health records, and production tracking (milk, eggs, etc.).
- **Workforce and Resource Management System**
Manages task assignment, attendance, salary records, and optimizes use of resources like water, fertilizer, and feed.
- **Agricultural Marketplace with QR-Based Verification**
Enables farmers to sell products directly to customers with QR-based traceability to ensure product authenticity and transparency.
- **Real-Time Notification and Alert System**
Provides instant alerts for weather updates, irrigation reminders, disease risks, and task notifications.

Overall, the system reduces manual work and improves data accuracy by integrating all farming operations into a single platform for efficient management and decision-making.

1.6 Aim and Objectives

1.6.1 Aim

The aim of this project is to develop a Smart Farm Management and Advisory System that improves agricultural productivity, operational efficiency, and decision-making through a centralized web-based platform for managing and monitoring farming activities. The system aims to modernize traditional farming practices by providing a structured and technology-driven digital solution that supports better farm coordination, resource management, and access to agricultural information for farmers and farm managers.

1.6.2 Objectives

- To develop a centralized farm management system
Integrate crop, livestock, workforce, and farm data into a single platform to improve coordination and efficient management of farm activities.
- To implement crop tracking and monitoring functionalities
Enable real-time tracking of crop growth stages and field operations to support better planning and farm supervision.
- To develop an AI-based advisory and disease detection system
Provide farming recommendations based on weather and farm data, while supporting early identification of crop diseases through image-based analysis.

- To implement livestock and workforce management features
Support basic livestock monitoring along with worker task allocation, attendance tracking, and coordination of farm labor.
- To develop a marketplace and notification system
Enable direct selling of farm products with QR-based transparency and provide timely alerts for farm activities, irrigation, and weather updates.

Chapter 2: Methodology

To efficiently develop the Smart Farm Management and Advisory System, we adopted the Agile Software Development Methodology. Agile provides a structured, iterative, and flexible approach that supports continuous improvement of the system through regular feedback. This methodology is suitable for this project because the system consists of multiple integrated modules such as farm management, livestock management, AI-based advisory, marketplace, and notification systems, which require continuous refinement and stakeholder involvement throughout the development lifecycle.

Agile methodology emphasizes incremental development, frequent testing, and early delivery of functional components, which aligns well with the complexity and real-time nature of agricultural operations.

The Agile Software Development Methodology adopted in this project is illustrated in Figure 1.

1. Requirement Gathering

The first phase involved identifying and understanding the challenges faced in traditional farm management systems. Requirements were gathered through direct discussions with the client (Annam integrated farm), along with literature study and analysis of real-world agricultural practices [1],[2].

The identified system requirements include the following functionalities:

- Centralized farm and livestock management system
- Crop lifecycle tracking from seed to harvest
- Task assignment and workforce management system
- Livestock feeding, health, and production tracking
- AI-based advisory and farming recommendation system
- disease detection system
- Agricultural marketplace with product listing and ordering
- QR-based product verification system

- Order and payment management system
- Notification and communication system for real-time updates

This phase helped establish a clear foundation for system development and ensured that all essential farm operations were included in the system scope.

2. System Design and Architecture

In this phase, the gathered requirements were translated into a structured system design.

- User-Centric Design: Defined different user roles such as Super Admin, Farm Manager, Farmers/Workers, Customers and Guest user with role-based access control.
- Wireframes & Prototypes: Designed user interfaces to ensure simplicity, responsiveness, and ease of use for all users.
- System Architecture: Designed a centralized web-based system integrating farm management, livestock management, AI advisory, marketplace, and notification modules into a single platform.

Technology Stack:

Frontend: React.js[3]

Backend: Django (Python)[4]

Database: PostgreSQL[5]

3. Development and Implementation

The system is developed using a modular approach, where each major functionality is implemented independently and later integrated into the main system.

Major modules include:

- Farm and livestock management module
- Crop monitoring and lifecycle tracking module
- Task and workforce management module
- AI-based advisory and risk management module
- Crop disease detection module
- Agricultural marketplace system

- QR-based product verification system
- Order and payment management system
- Notification and communication system

Development focuses on clean architecture, secure backend logic, responsive frontend design, and efficient API communication using Django REST Framework [4]. Version control is maintained using Git and GitHub for collaboration and source code management.

4. Testing and Quality Assurance

Testing is carried out throughout the development process to ensure system reliability and correctness.

- Unit Testing: Each module will be tested individually to verify functionality.
- Integration Testing: Ensures proper communication between frontend, backend, and database.
- System Testing: Validates the complete system workflow including farm operations and marketplace functions.
- User Acceptance Testing (UAT): Conducted with intended user (Annam integrated farm) to ensure the system meets real-world requirements [1].

Both manual and API testing tools (such as Postman) are used to ensure system stability and performance.

5. Deployment and Launch

After successful testing, the system will be deployed in a structured manner.

A staging environment is used for final verification before production deployment. The system is hosted on cloud infrastructure to ensure scalability, high availability, and global access. Frontend, backend, and database components are deployed and integrated in the production environment.

Proper documentation and user guidelines are provided to support system usage and onboarding.

6. Maintenance and Iterative Improvements

After deployment, the system is continuously monitored and improved to ensure long-term reliability.

System performance such as response time, uptime, and user activity is regularly analyzed. Bugs and issues are fixed through updates, and user feedback is collected for further improvements.

Future enhancements will include improved AI advisory accuracy, advanced analytics dashboards, and additional automation features to further improve system efficiency and scalability.

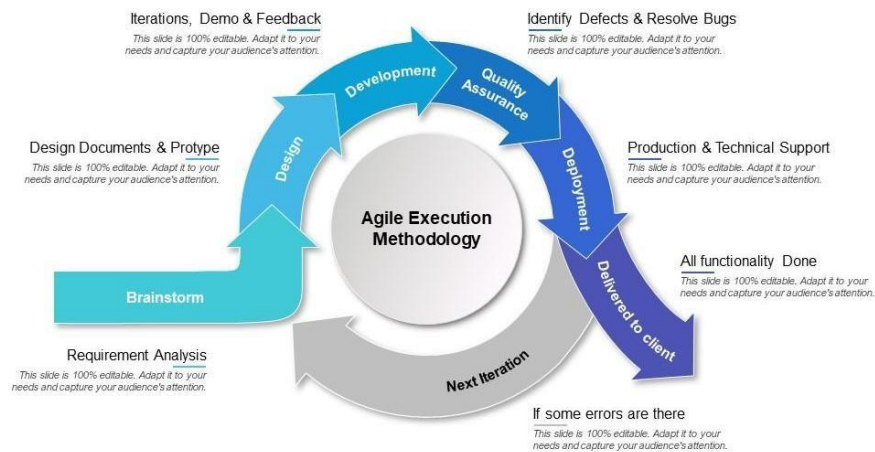


Figure 1 : Agile Methodology

Chapter 3: Requirements Identification

3.1 Functional and Non-functional requirements

3.1.1 Functional Requirements

1. Farm & Livestock Management System

- The system should manage farm records including fields, crop types, and planting details.
- The system should track crop from seed to harvest.
- The system should maintain centralized farm data and historical records for analysis.
- The system should manage livestock records including count, health, and production details.
- The system should monitor livestock health, feeding, and production (milk/eggs).
- Livestock management should be included as a supporting sub-module to assist overall farm operations.

2. Farm Operations & Task Management System

- The system should allow creation and assignment of farm tasks to workers.
- The system should track daily activities such as irrigation, feeding, and field work.
- The system should record crop and livestock updates by workers.
- The system should maintain task progress information such as pending, ongoing, and completed tasks, along with records of worker activities and task updates.
- The system should manage worker attendance records.
- The system should calculate and maintain worker salary information.

3. Crop Monitoring System

- The system should allow farmers and workers to update crop growth stages from seed to harvest.
- The system should monitor field conditions.
- The system should maintain crop progress history.
- The system should support real-time status updates.

4. AI-Based Advisory & Risk Management System

- The system should provide AI-based recommendations by integrating external weather APIs (such as OpenWeather) along with farm data collected from user inputs and system records.
- The system should support crop disease detection
- The system should generate alerts for weather risks using real-time weather API data, and for crop disease and environmental risks using predefined rules and AI-based image or data analysis from farm inputs.
- The system should provide irrigation and farming actions.

5. Agricultural Marketplace System

- The system should allow farmers to add products for sale.
- The system should enable customers to browse and purchase products.
- The system should support tracking.
- The system should manage product availability.

6. QR-Based Product Verification System

- The system should generate QR codes for products.
- The system should allow customers to verify product authenticity.
- The system should display product origin and farm details.
- The system should ensure product traceability.

7. Order & Payment Management System

- The system should process customer orders.
- The system should track order status.
- The system should support secure payments.
- The system should maintain invoices and payment history.

8. Notification & Communication System

- The system should send real-time notifications for farm activities.
- The system should notify task updates and assignments.
- The system should send alerts for crop, livestock, and risk conditions.
- The system should notify order and payment status.
- The system should support system-wide communication.

9. Reporting & Analytics System

- The system should generate crop and livestock performance reports.
- The system should provide productivity analytics and insights.
- The system should support dashboard-based decision-making.

10. System Administration & Configuration

- The system should manage users and roles.
- The system should configure AI, weather API, and QR settings.
- The system should monitor system performance and security.
- The system should manage logs and access control.

3.1.2 Nonfunctional requirements

1. Performance

The system should support multiple concurrent users with minimal delay. The system response time should remain within 2–3 seconds for normal operations, ensuring a smooth user experience even during peak usage.

2. Scalability

The system should support the expansion of farms, users, crops, and livestock without requiring major changes to the core architecture. It should also allow future integration of additional modules such as advanced AI features and analytics systems.

3. Usability

The system should provide a simple, responsive, and user-friendly interface that is accessible to users with basic technical knowledge. The interface should be optimized for both desktop and mobile devices to ensure ease of use.

4. Availability

The system should ensure at least 99% uptime and maintain continuous availability of services. Planned maintenance and unexpected failures should be handled with minimal downtime to ensure system reliability.

5. Security

The system should implement secure authentication for all users, ensuring only authorized access. All data transmission should be encrypted using SSL/TLS protocols. Role-based access control should restrict unauthorized access based on user roles. Passwords should be securely stored using hashing techniques to prevent data breaches.

3.2 System requirements

3.2.1 Hardware Requirements

1. Laptop or Desktop (Core i5 / Ryzen 5 or higher, 8GB RAM, SSD)
Used for system development, coding, database management, testing, and integration of frontend and backend modules.
2. User Devices (Desktop / Laptop / Smartphone with browser support)
Used by farmers, farm managers, customers, and administrators to access the web application through a browser.
Supports accessing dashboards, updating farm data, managing livestock, and using marketplace features.
3. Stable Internet Connection (4G/5G or Wi-Fi)
Required for real-time data updates, notifications, AI recommendations, and cloud communication.

3.2.2 Software Requirements

1. Operating System: Windows 10 or later / macOS
Used for development, testing, and system administration.
2. Web Browser: Chrome, Edge, or Brave
Required for accessing and interacting with the web application.
3. Code Editor: Visual Studio Code
Used as the primary development environment for frontend and backend coding.
4. Programming Language: Python 3.10 or above
Used for backend development and AI logic implementation in Django.

5. Frontend Runtime: Node.js 18 or above
Used for managing React.js frontend development and build processes.
6. Cloud Hosting Infrastructure (Vercel, Railway, Supabase)
Used to host frontend, backend, and database services.
Ensures scalability, high availability, and global access without physical server dependency.
7. Version Control: Git and GitHub
Used for version control, code management, and collaboration.
8. API Testing Tool: Postman
Used for testing and validating REST APIs between frontend and backend.
9. UI/UX Design Tool: Figma
Used for designing system interfaces and user experience.

3.2.3 Technologies

1. Frontend: React.js
 - Used to develop a dynamic, responsive, and component-based user interface.
 - Provides dashboards for farm management, livestock tracking, and marketplace operations.
2. Backend: Django (Python)
 - Handles business logic, authentication, and API services.
 - Manages farm data, livestock data, orders, and system operations securely.

3. Database: PostgreSQL

- Used for storing structured data such as farm records, livestock details, users, and transactions.
- Ensures data integrity and reliability.

4. API Layer: Django REST Framework

- Used to build RESTful APIs for communication between frontend and backend.
- Ensures smooth data exchange across system modules.

5. AI Integration: Google Gemini API

- Provides intelligent advisory and recommendation services..
- Used for crop recommendations, livestock guidance, and farming suggestions.

6. Weather Integration: OpenWeather API

- Provides real-time weather data for irrigation planning and risk analysis.
- Supports advisory and risk management decisions.

7. Version Control: Git and GitHub

- Used for source code management and collaborative development.

8. Cloud Deployment

- Vercel – used for frontend hosting
- Railway – used for backend deployment
- Supabase – used for PostgreSQL database hosting

- Ensures scalability, security, and continuous availability

9. AI/ML Models and Algorithms

- CNN Models (MobileNet)
Used for crop disease detection through image classification of leaf images.
- Rule-Based Logic System
Used for generating basic recommendations and alerts.

3.3 User Role

1. Super Admin

The Super Admin is the highest authority in the system who controls and monitors the entire Smart Farm Management and Advisory System.

- Manage all users and assign roles
- Monitor all farm activities across the system
- Configure system settings such as AI services, weather API, and QR system
- View reports and system analytics
- Manage system security and access control

2. Farm Manager

The Farm Manager is responsible for overall farm planning, coordination, and operational management.

- Create crop records including field, crop type, and planting date
- Assign tasks to farmers and workers
- Monitor crop stage updates reported by farmers
- Manage livestock records and health monitoring
- Handle irrigation planning and risk alerts
- Manage marketplace products, orders, and pricing
- Manage workforce and salary system

3. Farmers / Workers

Farmers and workers are responsible for daily farm operations and field level activities.

- Update crop stages from seed to harvest
- Perform daily farming activities in the field
- Report crop and livestock conditions
- Feed and care for livestock
- Record milk and egg production data
- Receive and complete assigned tasks
- Mark attendance

4. Customers

Customers are external users who interact with the system through the agricultural marketplace.

- View and purchase farm products
- Scan QR codes for product verification
- Place and track orders
- Make payments
- Provide feedback and ratings

5. Guest User

Guest Users are unregistered visitors who can access limited features of the system without an account.

- View platform information and services
- Browse products and marketplace listings
- View announcements and public content
- Access login and registration pages
- No access to orders, payments, or farm management features

Chapter 4: Project Plan

4.1 Gantt chart

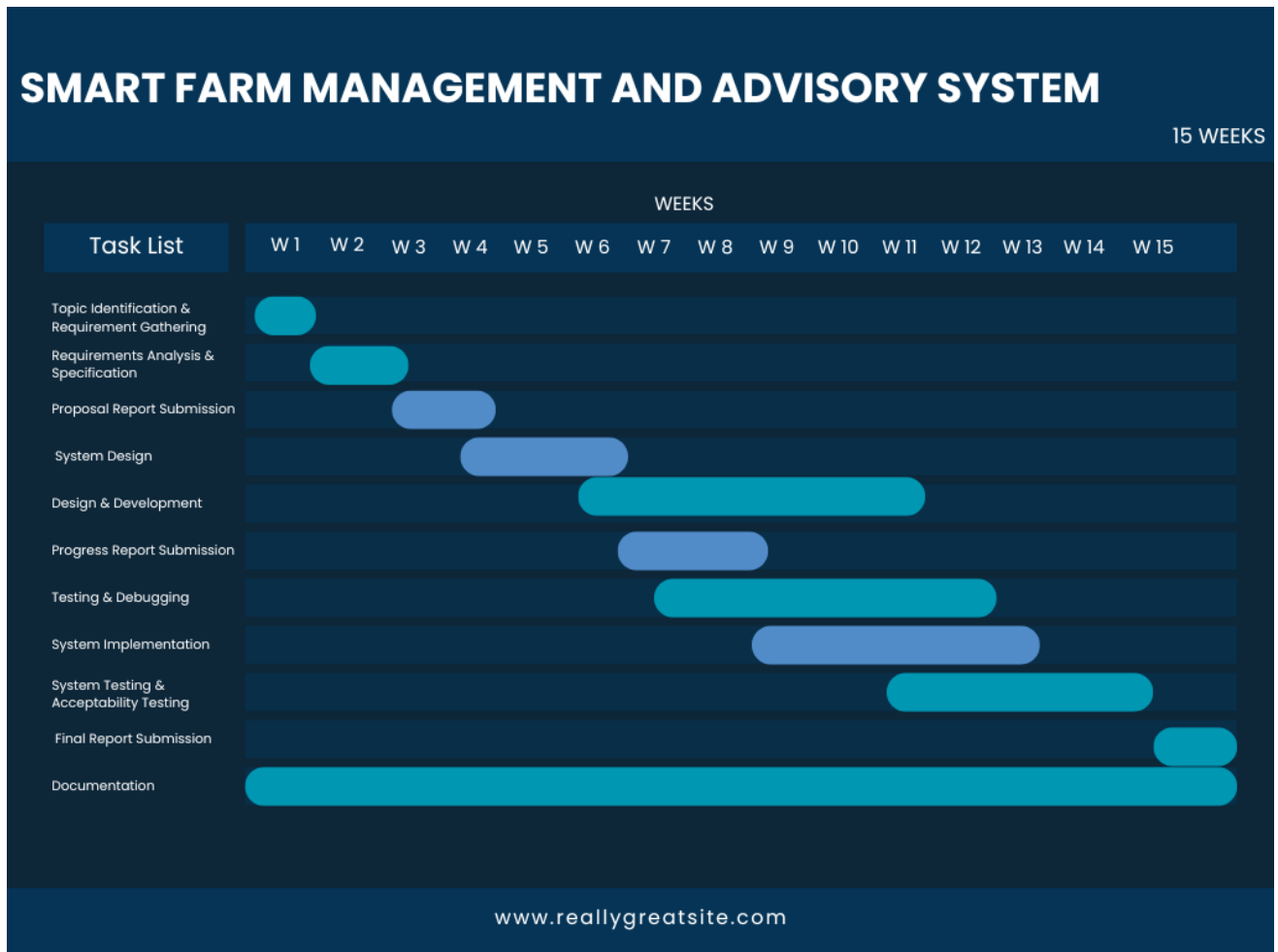


Figure 2 : Gantt Chart

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Appendixes

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28/04/2026

Chairman,
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Tharmapuram, Kilinochchi,
Sri Lanka.

Respectable Sir,

Requesting Support for Our Project | Smart Farm Management and Advisory System

We would like to introduce ourselves as a group of undergraduate students from Uva Wellassa University of Sri Lanka and are writing to kindly request your support for our academic project titled "Smart Farm Management and Advisory System."

As part of our degree program, we are undertaking this project with the aim of developing a web-based platform to improve farm management and decision-making. The system focuses on both crop and livestock management. It is designed to help farmers and farm managers manage farm activities, monitor production, and improve productivity through digital solutions.

The proposed system includes features such as AI-based advisory and forecasting, crop disease detection using image processing, irrigation and risk management, and an agricultural marketplace with product traceability.

We hope that your organization will support our project by providing insights, feedback, and if possible, sample data or real-world guidance to help us develop a practical and effective solution. Your support will greatly help us to improve the system and align it with real farming needs.

Should you have any suggestions or specific requirements, we would be happy to consider them in our project development. We believe this system can benefit both academic purposes and real-world agricultural practices.

We sincerely request your kind support and cooperation. We look forward to working with you on this initiative.

Thank You.

Yours sincerely,
S.kristo praveejiny,
R.Jathushan
R. Pirathijusha
K.Kavisana

Appendix A : Request Letter



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Tharmapuram, Kilinochchi, Sri Lanka

Chairman,
Annam Integrated Farms (Pvt) Ltd.
No. 565/12, Tharmapuram East
Tharmapuram, Kilinochchi,
Sri Lanka.

May 05, 2026

~~EST~~ ^{ITT} Students (Group IT),
Uva Wellassa University of Sri Lanka,
Badulla,
Sri Lanka.

Dear Students,

Subject: Confirmation of Project Collaboration – Intelligent Farming System for Crop and Livestock Optimization

We are pleased to officially confirm our collaboration on your undergraduate project titled "Intelligent Farming System for Crop and Livestock Optimization". We are genuinely excited about this opportunity to work alongside the student team from Uva Wellassa University.

We appreciate the effort, innovation, and commitment demonstrated by your group in developing a platform aimed at improving agricultural productivity and decision-making through technology.

The proposed system includes several innovative features such as:

- AI-based agricultural prediction engine.
- Smart farming advisory system using weather and soil data.
- Crop disease detection using image processing techniques.
- Smart irrigation and pest risk prediction system.
- Smart agricultural marketplace with product management and traceability support.

We believe this system will significantly help farmers by improving productivity, reducing risks, and increasing transparency in agricultural operations.

Appendix B: Confirmation Letter



Appendix C: Confirmation letter contd..